

platform where people can conduct experiments on their actual quantum hardware. Multinational companies such as JPMorgan Chase, Daimler (used in logistics for optimising vehicle delivery routes), Honda, Samsung and Barclays (settlement of large batches of transactions) were the first to sign up for the active testing of this service.

Google: Google's most recent quantum processor is called Willow. Unveiled in December 2024, this 105-qubit superconducting chip for the first time demonstrates exponential error suppression as qubit count scales.

Microsoft: In February this year, Microsoft announced the release of Majorana 1, the first quantum processing unit powered by topological qubits. The quantum computing development kit, the full quantum software stack, including the Q# programming language and the Quantum Development Kit (QDK) are all open sourced. Azure Quantum provides access to various quantum hardware and quantum-inspired services.

D-Wave Systems: D-Wave built the world's first commercially available quantum computer using simpler technology, which was limited in capability. The company is focusing on how quantum computing can be applied to artificial intelligence, machine learning, and difficult optimisation problems. In May 2025, it developed the 4400+ qubit quantum computing system called D-Wave Advantage2 to address cybersecurity and logistics optimisation. D-Wave's Leap Quantum LaunchPad gives companies a chance to explore the former's quantum-based technology for free for three months.

Rigetti Computing: Rigetti recently launched a 36-qubit multi-chip quantum processor. Its quantum computing system provides a full-stack solution that tightly integrates hardware and software, offering low-latency cloud access for hybrid quantum-classical workflows.

Intel: Intel has announced the delivery of a 49-qubit test quantum

processor called 'Tangle Lake' and a 12-qubit silicon chip called 'Tunnel Falls'.

Amazon Web Services: In February 2025, AWS launched its first quantum chip named Ocelot. This chip is still in the prototype stage.

NVIDIA: In March 2025, NVIDIA launched the NVIDIA Accelerated Quantum Research Center (NVAQC) in Boston. This centre is all about developing new quantum computing architectures and algorithms that can work together with NVIDIA's powerful AI supercomputers.

IonQ: IonQ has developed quantum computers based on trapped ion technology.

Other leading quantum computing vendors are Alibaba, PsiQuantum, Xanadu, QuEra Computing, Pasqal, and Riverlane.

The implications of true quantum computing at scale are staggering and can have an extraordinary impact on society. Quantum algorithms can be applied for:

- Optimisation problems such as scheduling and route planning to find the best possible solution from the many decisions or options that are available
- Search, sampling and pattern matching
- Quantum encryption

The various ways it can be used in different industries are listed in Table 3.

Challenges in quantum computing

Lack of good software: Only a handful of quantum algorithms (VQE, QAOA, Grover, Shor) demonstrate meaningful speedups today; most real-world problems still lack proven quantum advantage.

Table 3: Applications of quantum computing

Industry	Usage
Health care and pharma services	• Accelerating drug discovery • Drug design, molecular dynamics • Faster identification of candidate compounds and novel materials design • Optimising therapy treatment • Quicker time to market of new drugs
Financial services	• Trading strategies • Portfolio optimisation • Asset pricing and risk analysis • Detecting market instabilities
High tech	• Bidding strategies for advertisements • Online and product marketing • Software verification and validation
Transportation	• Self-driving cars • Air traffic management • Manage transportation requests • Goods delivery in short time
Energy	• Network design • Energy distribution • Oil well optimisation
Governments	• Quantum-safe encryption • National security and defence • Cryptography and cybersecurity • Secure communications • Supply chain resilience, logistics planning, and resource allocation across military and civilian agencies